

## **M8-Decision Tree**

### **Part 1**

#### **-Dependent Variable:**

Marital Status

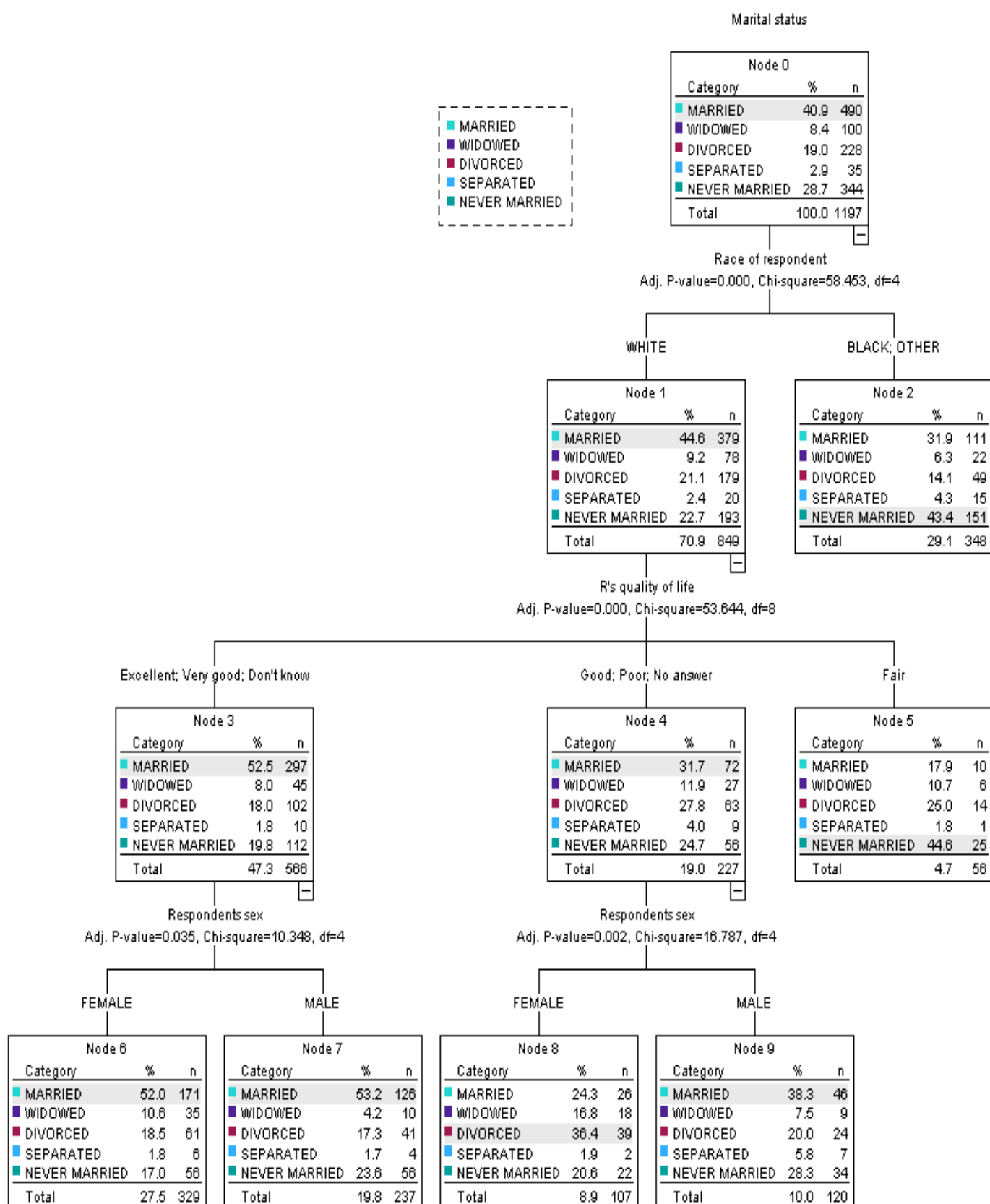
#### **-Independent Variable:**

Race/Sex/ Quality of list/ R's federal income tax/ Highest year of school completed

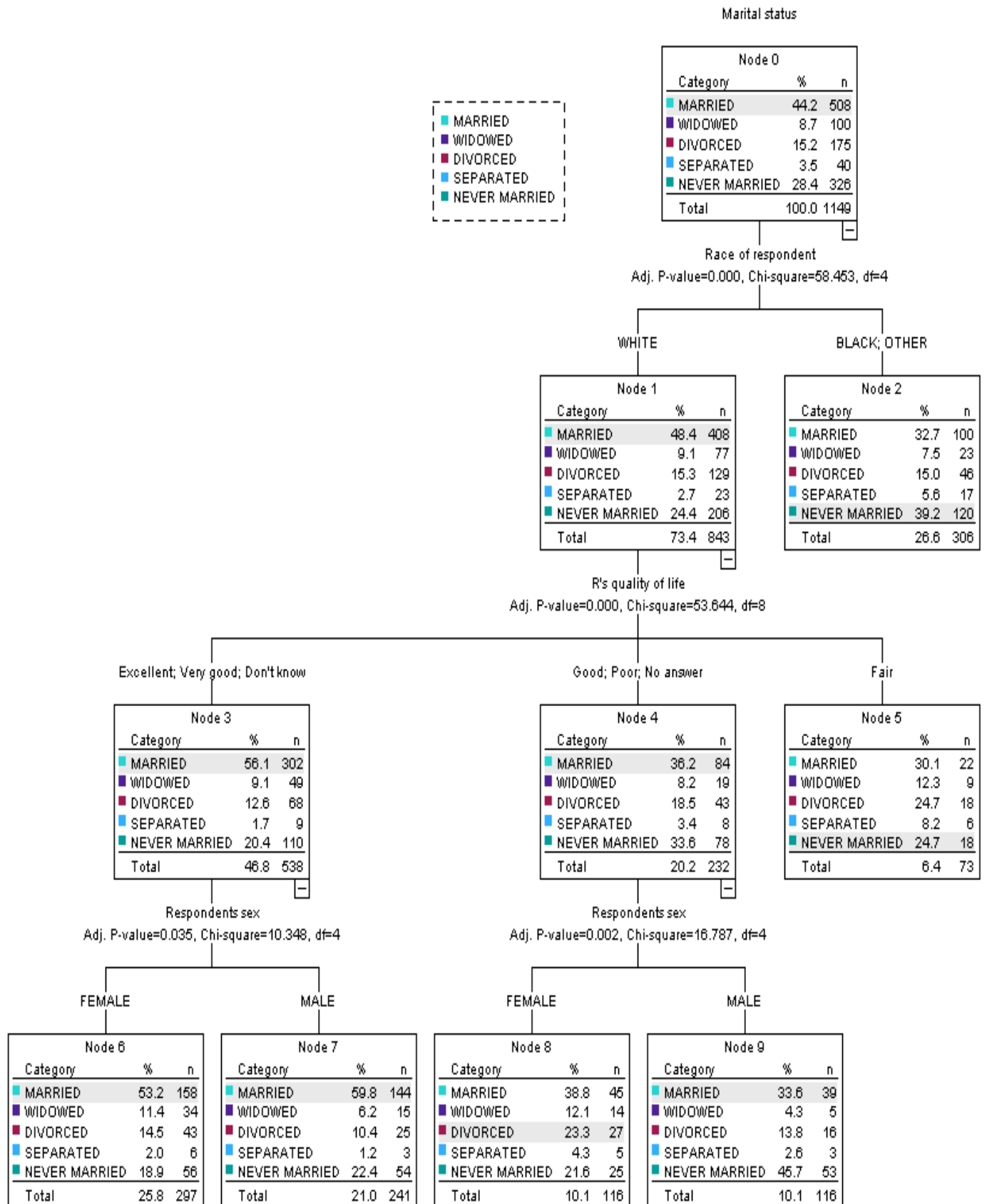
#### **- Purpose**

The purpose of this decision tree model was to predict marital status using five independent variables from the GSS2018 dataset, including race, sex, education level, quality of life, and federal income tax.

## Training Sample



## Test Sample



### **-Terminal Node 6 was selected for interpretation.**

This node included White female respondents reporting excellent or very good quality of life. In the Training sample, 52.0% of respondents in this node were married, compared to 53.2% in the Test sample. Since the percentages were very similar, the model does not appear to be overfit.

## **Part 2**

The purpose of this exercise was to create and validate a decision tree model predicting marital status using variables from the GSS2018 dataset. The model included race, sex, education level, quality of life, and federal income tax as independent variables. The decision tree showed that race was the strongest predictor, followed by quality of life and sex.

Terminal Node 6 consisted of White female respondents who reported an excellent or very good quality of life. In the Training sample, 52.0% of respondents in this node were married, compared to 53.2% in the Test sample. The similarity between these percentages suggests that the model generalized well to the Test sample and did not exhibit evidence of overfitting. In addition, the distributions of all other marital status categories were also similar between the Training and Test samples for Node 6, providing further evidence of model stability and consistency.

Overall, the decision tree identified meaningful predictors of marital status, including demographic characteristics, quality of life, income, and education. The similarity between the Training and Test sample results suggests that the model generalized well and demonstrated good predictive stability.